1. **How can the current distribution plan be improved? Formulate and solve an LP model to show the potential savings. Discuss how the current distribution plan and its cost change.**

Amount of malt shipped from plant i to brewery j in year 1 (1000 ton)				
	Istanbul	Ankara	Total	Capacity
Afyon	29.04	0.96	30.00	30.00
Konya	0	25.44	25.44	68.00
Import	0	0	0.00	20.00
Total malt	29.04	26.4		
Total beer	242	220	*(Million lt)*	

Amount of beer shipped from brewery j to distribution center k in year 1 (Million liters)									
	Istanbul	Izmir	Antalya	Bursa	Kayseri	Samsun	Export (Izmir)	Total	Capacity
Istanbul	103	74	0	60	0	0	5	242	250
Ankara	0	0	50	0	102	60	8	220.00	220
Total	103	74	50	60	102	60	13		
Demands	103	74	50	60	102	60	13		
								Total shipping cost	11.8006524731429
									(Million dollars)

Min Z = ∑i ∑j malt cost (i,j) * malt shipped (i,j) + ∑j ∑k beer cost (i,j) * beer shipped (j,k)

Where: i = malt plant, j = brewery, k = dist center

The total optimized shipping cost is now $11.8M, versus before it was $14.16M
Total savings is now $2.36M, or 16.7% less cost

Changes:

- Malt from Ayfton before was shipped almost entirely to Ankara and Import to Istanbul. The new plan now has Ayfton shipping to Istanbul, and Konya shipping to ankara
- With this, we eliminated all import malt since its shipping costs is more
- Beer distribution also changed to match distributors to closer breweries geographically

2. Is it cost-effective for Efes to import malt? If not, under what input parameter values would importing malt become a viable option?

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
C43	Afyon Istanbul	29.04	0	0.0264374	0.0026438	0.000822429
D43	Afyon Ankara	0.96	0	0.0166864	4.9E-05	0.0026438
C44	Konya Istanbul	0	0.010339	0.0368254	1E+30	0.010339
D44	Konya Ankara	25.44	0	0.0167354	0.0026438	4.9E-05
C45	Import Istanbul	0	0.002884145	0.0317784	1E+30	0.002884145
D45	Import Ankara	0	0.0142566	0.0325134	1E+30	0.0142566
C51	Istanbul Istanbul	103	0	0	0.031285714	1E+30
D51	Istanbul Izmir	74	0	0.040357143	1.56125E-17	1E+30
E51	Istanbul Antalya	0	0.013857143	0.051714286	1E+30	0.013857143
F51	Istanbul Bursa	60	0	0.017357143	0.008857143	1E+30
G51	Istanbul Kayseri	0	0.033428571	0.055142857	1E+30	0.033428571
H51	Istanbul Samsun	0	0.025071429	0.053	1E+30	0.025071429
I51	Istanbul Export (Izmir)	5	0	0.041857143	0.013857143	1.56125E-17
C52	Ankara Istanbul	0	0.031285714	0.032357143	1E+30	0.031285714
D52	Ankara Izmir	0	1.56125E-17	0.041428571	1E+30	1.56125E-17
E52	Ankara Antalya	50	0	0.038928571	0.013857143	1.62363E+12
F52	Ankara Bursa	0	0.008857143	0.027285714	1E+30	0.008857143
G52	Ankara Kayseri	102	0	0.022785714	0.033428571	1E+30
H52	Ankara Samsun	60	0	0.029	0.025071429	1E+30
I52	Ankara Export (Izmir)	8	0	0.042928571	1.56125E-17	0.013857143

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
C47	Total beer Istanbul	242	0.003178368	0	8	212
D47	Total beer Ankara	220	0.002008248	0	354.6666667	212
C53	Total Istanbul	103	0.003178368	103	8	103
D53	Total Izmir	74	0.043535511	74	8	74
E53	Total Antalya	50	0.041035511	50	8	5
F53	Total Bursa	60	0.020535511	60	8	60
G53	Total Kayseri	102	0.024892654	102	8	5
H53	Total Samsun	60	0.031106939	60	8	5
I53	Total Export (Izmir)	13	0.045035511	13	8	5
E43	Afyon Total	30	-4.9E-05	30	25.44	0.96
E44	Konya Total	25.44	0	68	1E+30	42.56
E45	Import Total	0	0	20	1E+30	20
J51	Istanbul Total	242	0	250	1E+30	8
J52	Ankara Total	220	-9.86914E-05	220	5	8

No it is not. We already produce enough domestically for the demand, we still have extra capacity in Konya if needed. And the cost to produce imported malt is much higher to ship, so while we are not overcapped, there is no need to import more malt at higher costs.

For importing to be viable, we would need to reduce the costs. Looking at the sensitivity report, we would need the cost to Istanbul to drop to .00288< per 1000 tons to be worth using, which would be a 9% reduction. For Ankara, we would need the price to drop to .01826, which is a much larger 44% reduction. Istanbul is the most viable option if the price is reduced.

3. **The linear programming model uses the current demand forecast for the next year to optimize the distribution plan. The forecasts are estimates and the actual demand could vary from the estimated value. Discuss the effects of beer demand variation on transportation costs using the information in the sensitivity report.**

    - The most expensive to serve is export/Izmir. This makes sense, as it's shipping beer to a port to export. If demand increased, it would go up by $45k
    - The cheapest to serve is Istanbul. This also makes sense, as the brewery in Istanbul has zero transport cost. Extra demand here would make price go up by about $3k
    - The allowable ranges tell you the range of demand you can be in without switching your routes. Interestingly, the allowable increase for each is 8, but the decrease varies heavily. This is because Istanbul has 8ML of slack capacity. Once demand goes beyond that, we would need to restructure.

4. **Analyze the situation taking into account the location(s) of new breweries,and locations of expansions of capacities and investment and transportation costs. Find the best decisions that minimize the total costs.**

Decision variables

Malt shipped x(i,j) (1000 tons)

From \ To	Istanbul	Ankara	Izmir	Sakarya	Adana	Samsun	IstNew	AnkNew	Row sum
Afyon	30.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	30.00
Konya	0.00	26.40	0.00	0.00	0.00	10.80	0.00	5.40	42.60
Import (Izmir)	0.00	0.00	14.40	0.00	0.00	0.00	0.00	0.00	14.40

Binary decisions

Site	Open	Expand
Izmir	1	1
Sakarya	0	0
Adana	0	0
Samsun	1	0
IstNew	0	0
AnkNew	1	0

Beer shipped y(j,k) (million liters)

From \ To	Istanbul	Izmir	Antalya	Bursa	Kayseri	Samsun	Export (Izmir)	Row sum
Istanbul	150.00	0.00	0.00	100.00	0.00	0.00	0.00	250.00
Ankara	0.00	20.00	60.00	0.00	140.00	0.00	0.00	220.00
Izmir	0.00	90.00	0.00	0.00	0.00	0.00	30.00	120.00
Sakarya	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Adana	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Samsun	0.00	0.00	0.00	0.00	0.00	90.00	0.00	90.00
IstNew	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
AnkNew	0.00	0.00	40.00	5.00	0.00	0.00	0.00	45.00
Col sum (supply to DC)	150.00	110.00	100.00	105.00	140.00	90.00	30.00	
Demand	150.00	110.00	100.00	105.00	140.00	90.00	30.00	

CAPACITY:

Annual capacity of malt plants	*(1000 tons/year)*
Afyon	30
Konya	68
Import (Izmir)	20

Checks

Yield (million liters per 1000 tons malt) 8.333333

Brewery	Malt in (1000 tons)	Beer available (=Yield*Malt)	Beer shipped (row sum)	Capacity
Istanbul	30.00	250.00	250.00	250.00
Ankara	26.40	220.00	220.00	220.00
Izmir	14.40	120.00	120.00	120.00
Sakarya	0.00	0.00	0.00	0.00
Adana	0.00	0.00	0.00	0.00
Samsun	10.80	90.00	90.00	90.00
IstNew	0.00	0.00	0.00	0.00
AnkNew	5.40	45.00	45.00	90.00

Objective (Minimize total cost, million $)	
Malt transport cost	1.7407
Beer transport cost	9.8286
Fixed investment cost	12.0000
TOTAL COST	23.5692

A mixed-integer model was solved to determine the optimal locations for new breweries and capacity expansions while minimizing total annual cost (investment + transportation), subject to demand, supply, and capacity constraints.

The optimal solution opens Izmir (with expansion to 120 capacity), Samsun (90 capacity), and AnkNew (90 capacity). Sakarya, Adana, and IstNew are not open. Existing breweries in Istanbul (250) and Ankara (220) remain in operation.

The resulting minimum total annual cost is:

23.569 million dollars, composed of:

- 12.000 million in fixed investment cost
- 1.741 million in malt transportation cost
- 9.829 million in beer transportation cost

The solution reflects a balance between investment and logistics efficiency. Expanding Izmir reduces western and export transport costs, while opening Samsun and AnkNew reduces long-haul shipments and capacity pressure in the network.

5. **What are the effects of some different (non-optimal) scenarios in terms of locations of new breweries and expansions? It may be useful for the management to know the effects of different policies. They may appreciate seeing different solutions that yield slightly higher costs as they may have other advantages that have not been considered in the model.**

Checks				
Yield (million liters per 1000 tons malt)			8.333333	
Brewery	**Malt in (1000 tons)**	**Beer available (=Yield*Malt)**	**Beer shipped (row sum)**	**Capacity**
stanbul	24.60	205.00	205.00	250.00
Ankara	26.40	220.00	220.00	220.00
zmir	14.40	120.00	120.00	120.00
Sakarya	10.80	90.00	90.00	90.00
Adana	0.00	0.00	0.00	0.00
Samsun	0.00	0.00	0.00	0.00
stNew	0.00	0.00	0.00	0.00
AnkNew	10.80	90.00	90.00	90.00
Objective (Minimize total cost, million $)				
Malt transport cost	1.5679			
Beer transport cost	11.9868			
Fixed investment cost	13.5000			
TOTAL COST	**27.0547**			

*example total cost for the non optimal scenario when we remove Samsun

In our excel file, for each new scenario, we made a copy of the model and ran the solver with the new constraint(s) added. When we did this, we found these insights as well as the exact cost increases shown in the slides:

1) Fewer new breweries
If management avoids opening one of the recommended new sites, fixed investment falls, but beer must be shipped longer distances from Istanbul/Ankara/Izmir. This increases transportation cost and makes the system more sensitive to demand changes and capacity bottlenecks.

2) No expansion at Izmir
Skipping the Izmir expansion reduces investment but shifts more volume to be served from other breweries, increasing beer transportation cost, especially for the Izmir market and export lane. This policy is attractive if export demand is uncertain, but it usually raises recurring logistics costs.

3) Alternative site choice
Opening a different candidate site may be slightly more expensive but could be preferred if management values proximity to specific customers, future growth corridors, labor availability, or regional risk diversification. The trade-off is higher modeled cost in exchange for non-modeled benefits.

4) Centralized operations
Limiting the number of operating breweries simplifies management and operations, but generally increases shipping distances and total transport cost. In some cases, a highly centralized policy can even become infeasible if total available capacity is insufficient to meet all demand.

6. **What is the effect of operating the breweries at slightly lower capacities than full capacities? There could be advantages to not forcing full capacity utilizations.**

Checks				
Yield (million liters per 1000 tons malt)			8.333333	
Brewery	**Malt in (1000 tons)**	**Beer available (=Yield*Malt)**	**Beer shipped (row sum)**	**Capacity**
Istanbul	27.00	225.00	225.00	225.00
Ankara	23.76	198.00	198.00	198.00
Izmir	14.04	117.00	117.00	117.00
Sakarya	0.00	0.00	0.00	0.00
Adana	0.00	0.00	0.00	0.00
Samsun	11.40	95.00	95.00	117.00
IstNew	0.00	0.00	0.00	0.00
AnkNew	10.80	90.00	90.00	90.00
Objective (Minimize total cost, million $)				
Malt transport cost	1.7257			
Beer transport cost	10.2471			
Fixed investment cost	12.5000			
TOTAL COST	**24.4729**			

*example total cost when capacity is at 90%. Repeated for 80-95% capacity

Capacity %	Total Cost	Change
95	23.81M	+.24M
90	24.47M	+ .90M

85	25.27M	+1.70M
80	27.16M	+3.59M

Recommendation and takeaways from producing below capacity:

When we reduce capacity by certain amounts, we see increasing costs in return. Once capacity goes below 90% capacity, the total costs accelerates sharply as the model needs to open additional breweries. Because of this, we would want to stay above 90% if we had to stay below full capacity to avoid any extra costs. Of course 100% capacity is the most optimal way to minimize costs, but in a scenario where that's possible, staying above the 90% threshold would be ideal.